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PMC12293755 | Miyaura 硼化 / C – H 硼化 | 采集状态 imported_unchecked

标题	Synthesis and Cytotoxicity Evaluation of Denitroaristolochic Acids: Structural Insights and Mechanistic Implications in Nephrotoxicity
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出处	Biomolecules 2025
DOI	10.3390/biom15071014
原文链接	https://pmc.ncbi.nlm.nih.gov/articles/PMC12293755/
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摘要（原文节选）

The efficient synthetic routes and evaluates cytotoxic profiles of denitroaristolochic acids II – V (DAA-II – V) were demonstrated in this study. Based on retrosynthetic analysis, a modular synthetic strategy was developed through Suzuki – Miyaura coupling, Wittig reaction, and bismuth triflate-catalyzed intramolecular Friedel – Crafts cyclization to efficiently construct the phenanthrene core. Process optimization significantly improved yields: aryl bromide intermediate A reached 50.8% yield via bromination refinement, wh ⋯

数据表 · Table 2（9 行，含数值行 9 行）

Optimization of reaction conditions for compound 2.

Entry	Reagent	Catalyst	Solvent	Time	Temperature	Yield
1	NBS (1.1 eq)	m-CPBA (0.1 eq)	methanol	12 h	r.t./65 ° C	—
2	NBS (1.1 eq)	m-CPBA (0.1 eq)	DCM	12 h	r.t./40 ° C	—
3	NBS (1.1 eq)	m-CPBA (0.1 eq)	THF	12 h	r.t./66 ° C	—
4	dibromohydantoin (1.1 eq)	m-CPBA (0.1 eq)	methanol	12 h	r.t./65 ° C	—
5	dibromohydantoin (1.1 eq)	m-CPBA (0.1 eq)	DCM	12 h	r.t./65 ° C	—
6	dibromohydantoin (1.1 eq)	m-CPBA (0.1 eq)	THF	12 h	r.t./40 ° C	—
7	dibromohydantoin (1.1 eq)	/	acetic acid	12 h	r.t.	—
8	NBS (1.1 eq)	/	acetic acid	12 h	r.t.	—
9	Br2 (1.1 eq)	/	acetic acid	6 h	r.t.	50.8%

正文对应句（逐字）：The isolation and characterization of AA derivatives achieved significant progress in 1984 when Basudeb Achari ’ s research group4 first isolated denitroaristolochic acid IV (DAA-IV) from Aristolochia indica [26], providing conclusive evidence for the natural occurrence of this class of compounds. Sub

数据表 · Table 3（10 行，含数值行 10 行）

Optimization of reaction conditions for carboxyl group hydrolysis.

Entry	Reagent	pH	Solvent	Time	Temperature	Yield
1	H2SO4	0.5	THF/DCM	overnight	Reflux	<10%
2	H2SO4	1.5	THF/DCM	overnight	Reflux	<10%
3	H2SO4	2.5	THF/DCM	overnight	Reflux	<10%
4	HCl	0.5	THF/DCM	overnight	Reflux	<10%
5	HCl	1.5	THF/DCM	overnight	Reflux	<10%
6	HCl	2.5	THF/DCM	overnight	Reflux	<10%
7	H3PO4	0.5	THF/DCM	overnight	Reflux	<10%
8	H3PO4	1.5	THF/DCM	overnight	Reflux	<10%
9	H3PO4	2.5	THF/DCM	overnight	Reflux	<10%
10	H3PO4	1.5	diphenyl ether	1 h	260 ° C	——

正文对应句（逐字）：The isolation and characterization of AA derivatives achieved significant progress in 1984 when Basudeb Achari ’ s research group4 first isolated denitroaristolochic acid IV (DAA-IV) from Aristolochia indica [26], providing conclusive evidence for the natural occurrence of this class of compounds. Sub

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